DeepLearning For Medical Image

Swarup G L
Roll NO – AD23B1053

Breast Ultrasound Classification Using VGG16

1. Objective and Dataset

The goal of this project is to classify breast ultrasound scans into three diagnostic categories: **Normal**, **Benign**, and **Malignant**.

The data used is the **Breast Ultrasound Images Dataset (BUSI)**, which contains 780 scans. Because the dataset folders include both the raw scans and radiologist-drawn segmentation masks (files ending in _mask.png), a custom data loader was written to strictly filter out the masks. This ensures the model learns to diagnose based purely on the raw ultrasound images.

2. Model & Transfer Learning

This project uses **Transfer Learning** with the **VGG16** neural network architecture.

- **Why ImageNet:** We initialized the model using standard **ImageNet** weights. Because the model has already learned to detect sharp edges, contrast, and complex textures from over a million everyday images, it serves as an excellent foundational "eye" for detecting the jagged edges and shadows of tumors.
- **Why not RadImageNet:** While RadImageNet is a highly specialized medical dataset, its pre-trained weights are primarily designed for newer models like ResNet50. Since we specifically wanted to use VGG16, standard ImageNet weights provided the most reliable and natively supported baseline.
- **Training Strategy:** The foundational layers of the VGG16 model were frozen, and only the final classification layer was retrained to output our 3 specific medical classes.

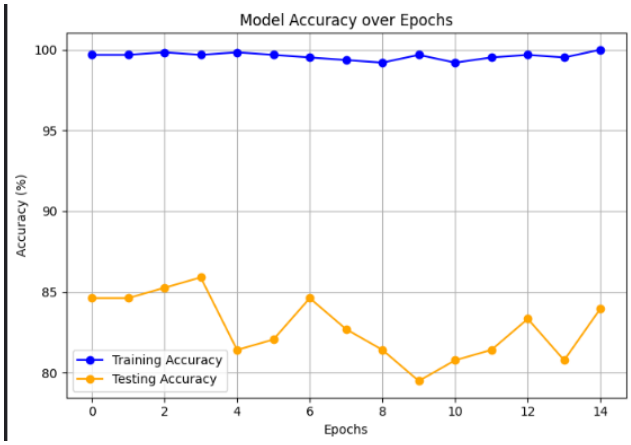
3. Data Processing and Splitting

- **Preprocessing:** All images were resized to 224x224 pixels and mathematically normalized to match the VGG16 input requirements.
- **Data Split:** The 780 images were randomly divided into an **60% 20% Training Set** and a **20% Testing Set**. The random seed was locked to ensure these splits are perfectly reproducible every time the code is run.

4. Results and Visualizations

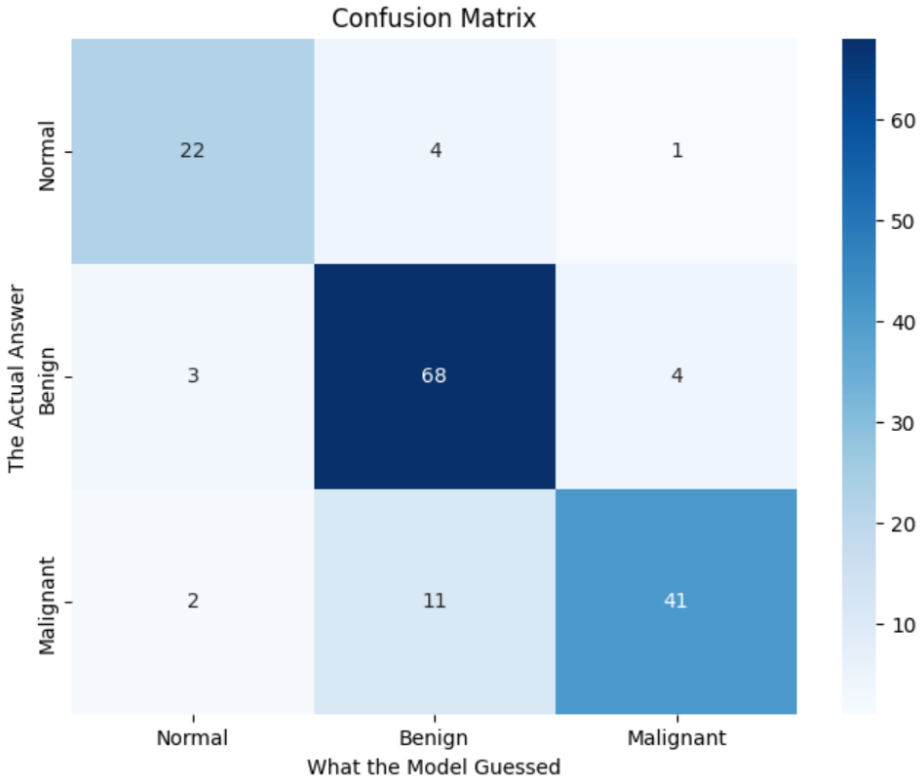
Model Accuracy and Loss Over Time

The graphs below track how the model learned over 15 epochs. A successful training run shows the Loss (error) decreasing and the Accuracy increasing for both the training and testing data.



Confusion Matrix

This heatmap shows exactly where the model succeeded and where it got confused on the unseen test data. For example, it reveals if the model accidentally classified a truly "Malignant" tumor as "Benign."



Thank You